

Hermes F&O Trading Master Guide - FINAL VERIFIED EDITION

Integrated review of the supplied 11-page quant/microstructure document plus current NSE/SEBI material and academic research. Unsupported or overbroad claims are explicitly qualified.

Risk and scope: This is a research/training manual. It does not provide guaranteed-profit methods or individualized investment advice. Futures and options are leveraged products; model risk, execution risk, gap risk and tail risk can exceed backtest expectations.

What I Verified and Changed

Item from supplied PDF	Verification result	Final treatment
SEBI FY26: 87.7% lost; ~Rs 91,685 crore aggregate loss	Verified against SEBI's Aug 20, 2026 study listing and contemporaneous reporting of the study.	Retained with fiscal-year scope.
FY22-FY26 continuing cohort: 0.5% profitable all five years; 65.6% loss-making every year	Supported by reporting of the new five-year cohort analysis.	Retained with cohort qualifier.
99% of FPI/proprietary profits from algo entities	Reported from SEBI's FY26 study; SEBI's algo-entity definition is classification-based.	Retained, explicitly saying this is not causal proof.
FY24 97% FPI / 96% proprietary algo-profit share	Historical SEBI finding, not a FY26 number.	Retained only as a separate FY24 comparison.
NSE equity-derivatives algo share ~70% in FY25	NSE Market Pulse supports the figure as a mode-of-trading turnover share.	Retained with FY25 and turnover context.
OFI $R^2 > 65\%$ as a general fact	Not justified as a universal statistic. Cont et al. establish a robust relation, not one universal R^2 .	Removed as universal; replaced with source- and sample-dependent wording.
Fractional Kelly 10-30% preserves 75-90% growth / cuts variance >70%	Not a universal result; depends on distribution and estimation error.	Removed as a fixed empirical promise. Use fractional Kelly only after calibration; conservative fixed-risk sizing is baseline.
CVaR $\leq 1.75\%$, leverage $\leq 4x$, daily stop 2% etc.	These were presented as institutional thresholds without evidence that they are universal standards.	Removed from 'verified' rules; replaced with illustrative policy-design examples.
VPIN >95th percentile predicts flash crashes	Empirically disputed in the literature.	Retained only as a toxicity/liquidity-research feature with controversy warning.
Footprint 300% / 3 stacked imbalances = institutional initiation	Trading heuristic, not a verified universal law.	Retained as a hypothesis to test, not a factual trigger.
Value Area = one standard deviation / 68-70% volume	Platform convention, not universal AMT law.	Rewritten as a common volume-profile convention.
Dealer GEX can be directly inferred from public OI	Dealer side/position is not directly observable from aggregate OI alone.	GEX is retained as an estimate with explicit assumptions.
DSR >0.95 required for live capital	0.95 can be a chosen policy threshold, not a universal law.	Retained as an optional internal gate, not an academic requirement.
Sub-microsecond hardware SLA for Hermes	High-frequency institutional infrastructure, not a requirement for retail API trading.	Moved to an HFT reference note; excluded from Hermes V1 requirements.

1. F&O Reality and the Objective

- SEBI's August 20, 2026 study reports 87.7% of individual equity-derivatives traders incurred losses in FY26, with aggregate individual net losses around Rs 91,685 crore. The same study reports a five-year continuing cohort in which only 0.5% were profitable in all five years and 65.6% lost money in every year.

- The FY26 study also reported that 99% of the profits of FPIs and proprietary traders were made by 'Algo entities'. The term is a regulatory classification: an entity that executed at least one trade using an algorithmic order. This should not be interpreted as proof that algorithms caused the profits.
- Earlier SEBI FY22-FY24 research reported 97% of FPI profits and 96% of proprietary-trader profits from algo entities. Keep this historical result separate from FY26.
- NSE Market Pulse reported about 70% of equity-derivatives trading by algorithmic mode in FY25. That is a turnover/trading-mode statistic, not the same thing as SEBI's profit-attribution measure.
- Success should be defined by net expectancy after realistic costs, controlled drawdown, robust out-of-sample performance and survival through regime changes.

2. Core Mechanics and Pricing

- Master spot, futures, calls, puts, moneyness, strike, expiry, premium, intrinsic/extrinsic value, lot size, margin, leverage, settlement, rollover and basis.
- Master Delta, Gamma, Theta, Vega, Rho, plus higher-order Greeks such as Vanna, Volga, Charm and Color as the system matures.
- Black-Scholes baseline: $C = S \cdot N(d1) - K \cdot \exp(-r \cdot t) \cdot N(d2)$; $d1 = [\ln(S/K) + (r + \sigma^2/2)t] / (\sigma \cdot \sqrt{t})$; $d2 = d1 - \sigma \cdot \sqrt{t}$.
- Heston: stochastic-volatility model that can reproduce dynamics unavailable to constant-volatility Black-Scholes.
- Monte Carlo: simulate outcome paths to estimate range, target/stop probabilities and tail exposure. Never confuse a simulated probability with a guaranteed forecast.

3. Current NSE Contract Setup

- NSE's current specification, updated August 11, 2026, lists NIFTY 50 index futures on a three-month cycle and NIFTY 50 index options with weekly/monthly/longer-dated expiries as specified by the exchange. The published expiry day for NIFTY 50 is Tuesday, shifting to the previous trading day if Tuesday is a trading holiday.
- Hermes should ingest current exchange contract metadata, strike intervals, expiry, lot-size and trading-status information instead of relying on hard-coded historical assumptions.

4. Classical Trading Algorithms

- Trend: EMA/SMA alignment, multi-horizon momentum and VWAP slope/reclaim are useful state variables.
- Breakout: range break + independently tested confirmation such as volume, ATR expansion, VWAP alignment and futures confirmation.
- Pullback: established trend -> controlled retracement -> structural reclaim. Fibonacci can be a feature, never a required truth.
- Mean reversion: $Z = (P - \mu) / \sigma$; only when regime evidence supports range behaviour.
- Time-series momentum: $\text{signal} = \text{sign}(P_t - P_{(t-N)})$. Academic evidence supports a time-series-momentum phenomenon across futures markets at medium horizons, but it does not guarantee the same edge in every intraday market.
- ATR: use volatility-scaled stops and risk sizing rather than fixed point distances.

5. Order Flow Imbalance (OFI) - Deep Quant Addition

- At the top of the book, OFI can be expressed using event-by-event changes in bid and ask queues. A convenient form is $\text{OFI}_n = \Delta Q_{\text{bid},n} - \Delta Q_{\text{ask},n}$, where queue changes account for price moves, additions, executions and cancellations.
- Cont, Kukanov and Stoikov found a robust short-interval relation between price changes and OFI in NYSE data, with price impact inversely related to market depth. The result is sample-dependent and should not be converted into a universal R^2 or universal predictive coefficient.

- Multi-Level OFI (MLOFI): aggregate OFI across several price levels with weights that can decrease with depth. This is a strong candidate feature for Hermes when reliable L2 data are available.
- Recommended Hermes features: OFI, MLOFI, spread, queue imbalance, order-add/cancel rates, aggressive trade imbalance, replenishment rate and time since last price move.

6. Stoikov Microprice and Book-State Modeling

- The ordinary midpoint $M = (\text{Ask} + \text{Bid}) / 2$ ignores queue asymmetry. A simple queue-weighted microprice is $\text{Microprice} = (\text{Ask} * \text{BidSize} + \text{Bid} * \text{AskSize}) / (\text{BidSize} + \text{AskSize})$.
- More advanced Stoikov-style models treat book imbalance and spread as a Markov state and estimate the expected future mid-price conditional on that state. The practical lesson is more important than the exact notation: book state changes are predictive features, not certainties.
- For Hermes, first test the simple microprice and imbalance. Add Markov/queue-state models only after establishing incremental out-of-sample value over a simple baseline.

7. VPIN / Flow Toxicity - Use with a Research Warning

- VPIN estimates order-flow imbalance in volume time. It is a candidate measure of flow toxicity and liquidity stress.
- The literature contains an explicit dispute over whether VPIN reliably predicts crashes or volatility better than standard indicators. Therefore Hermes should treat VPIN as one feature among many, and test it under multiple trade-classification and sampling specifications.
- Do not encode 'VPIN > 95th percentile = flash crash' as a deterministic rule. A safer use is: unusually high toxicity + widening spread + collapsing depth + volatility jump -> reduce liquidity provision / reduce new risk.

8. Footprint, CVD, Absorption and Auction-Market Structure

- Delta: Ask-traded volume minus Bid-traded volume. CVD: cumulative sum of bar-level delta.
- Absorption: large aggressive volume at an extreme without corresponding price extension can indicate passive liquidity absorbing flow. This is a hypothesis that needs confirmation, not a guarantee of reversal.
- Footprint divergences: price makes a new extreme while CVD fails to confirm. Treat as a context signal and test false-positive rates.
- Diagonal imbalances: large ask/bid ratios at adjacent prices can be useful for feature engineering. Specific thresholds such as 300% and 'three stacked levels = institutional initiation' are trading heuristics, not established universal laws.
- Auction Market Theory / Volume Profile: POC, HVN and LVN can provide structural context. Value Area definitions vary by platform/method; the common 68-70% convention should not be described as a universal one-standard-deviation law.

9. GEX, Gamma Flip, Vanna and Charm

- Dealer hedging can create feedback between option positions and underlying trading. In positive gamma conditions, hedging can tend to be counter-cyclical; in negative gamma conditions, it can tend to be pro-cyclical.
- Critical limitation: aggregate public OI does not directly reveal dealer net positions. Any GEX estimate requires assumptions about participant positioning, sign conventions, multipliers and which options are effectively dealer risk.
- Use GEX as an estimated regime feature. A 'Gamma Flip' is a model-derived price level where the estimated aggregate gamma changes sign; do not treat it as an objectively observed market wall.
- Vanna ($d\Delta/d\sigma$) and Charm ($d\Delta/dt$) can be added later to explain how hedging needs may change as implied volatility and time-to-expiry change.

10. Volatility, IV-RV and Surface Modeling

- Use IV(K,T) rather than a single IV. Track ATM IV, skew, smile, term structure, changes in surface slope and unusual cross-expiry distortions.
- Compare implied volatility with realized/forecast volatility. Indian options research found volatility-risk-premium information but also found strategy returns can fall materially once normal transaction costs are included.
- High IV is not automatically a short-volatility signal. Require favorable forecast RV, acceptable skew/tail risk, manageable event exposure and sufficiently liquid execution.
- Heston/SABR/Monte Carlo belong in the pricing/risk layer; a machine-learning model should learn residual information rather than blindly replace the entire pricing framework.

11. Statistical Arbitrage, OU and Kalman Filters

- For economically linked assets, define $\text{Spread} = Y - \beta X$. Test stationarity and cointegration before trading.
- An Ornstein-Uhlenbeck model $dZ = \theta(\mu - Z)dt + \sigma dW$ can estimate mean-reversion speed. Half-life = $\ln(2)/\theta$ only under the chosen OU parameterization.
- Use a Kalman filter to estimate a time-varying hedge ratio β_t and a dynamic spread mean/variance.
- Candidate rule: enter when absolute Z is sufficiently extreme, exit near equilibrium, and apply a structural stop if cointegration breaks. Entry/exit thresholds must be learned from out-of-sample data rather than imported as universal constants.

12. DeepLOB and High-Frequency ML

- The supplied PDF's DeepLOB section correctly identifies the architecture family: convolutional feature extraction + Inception-style multi-scale processing + LSTM temporal modelling.
- The original DeepLOB paper reports an F1 score around 83.4% on the FI-2010 benchmark at $k=10$. This is a benchmark result, not an NSE profitability claim. A current reproduction/search result confirms the original benchmark figure, while also illustrating that reproductions can perform materially worse.
- For Hermes, use 40-feature LOB snapshots across 10 bid/ask levels only when that exact data is available and correctly normalized. Start with logistic/XGBoost baselines and compare incremental performance.
- Target probabilities of Up/Flat/Down or thresholded future movement, not exact future prices. Evaluate macro-F1, calibration, precision by class and, most importantly, net trading performance after spread/latency costs.

13. Backtest Validation: Purged/Embargoed + CPCV + DSR

- Walk-forward: preferred practical baseline for non-stationary time series. It is not a magic cure for overfitting.
- Purged cross-validation: remove overlapping training observations whose labels overlap the test period.
- Embargo: leave a gap after test observations to reduce leakage from temporal dependence.
- CPCV: Combinatorial Purged Cross-Validation creates multiple out-of-sample paths and is useful when a research program has many candidate configurations.
- DSR: Deflated Sharpe Ratio adjusts for selection bias/multiple testing and non-normal returns. Do not hard-code $\text{DSR} > 0.95$ as a universal academic rule; make the approval threshold an explicit Hermes governance policy.
- Research ledger: record every model, parameter set, feature set, experiment and test result so that the number of trials is known.

14. Risk Engine: VaR, CVaR and SPAN Awareness

- Expected Shortfall/CVaR: estimate the average loss in the tail beyond a chosen VaR quantile. It is useful because option returns can be skewed and fat-tailed.

- Do not copy arbitrary institutional limits. The supplied document's example values (e.g. CVaR $\leq 1.75\%$, 4x futures leverage, 2% daily stop) are policy choices, not universal standards.
- NSE Clearing uses SPAN as a portfolio-based margin/risk system and scenario-tests price and volatility changes. Hermes should ingest current margin/risk files where available and treat broker/exchange margin requirements as external constraints.
- Hard controls should exist outside ML: maximum daily loss, maximum trade loss, maximum notional, concentration, gamma/vega exposure, slippage and drawdown.

15. Position Sizing

- Baseline: $\text{PositionSize} = \text{MaxRiskPerTrade} / (\text{StopDistance} * \text{ValuePerPoint})$.
- Use fixed fractional sizing first. Fractional Kelly can be explored after probability calibration is robust; do not promise a fixed percentage of Kelly will preserve a fixed percentage of growth or variance reduction.
- Use structural, volatility, time, probability and information stops. Never use martingale or uncontrolled averaging down.
- For options, size against scenario P&L and Greeks, not only premium paid.

16. Execution Algorithms

- TWAP: time-sliced execution. VWAP: volume-aware execution. POV: target participation rate. Iceberg: partial display where supported.
- Almgren-Chriss: formalizes the trade-off between market impact and risk of slow execution.
- Avellaneda-Stoikov: an advanced market-making framework for inventory-sensitive spreads. It is a research/market-making model, not necessary for directional Hermes V1.
- The supplied PDF listed microsecond-level FPGA/co-location SLAs. Those describe an institutional HFT architecture and should not be presented as requirements for a retail API system. Hermes V1 should optimize correctness, reliability, broker compliance and measured slippage first.

17. Indian Retail Algo Compliance - Architecture Impact

- NSE's published retail-algo framework sets an initial Threshold Orders Per Second at not more than 10 orders per second per exchange/segment; exchange/broker thresholds and rules can change.
- Client-developed algos above the applicable threshold require exchange registration through the broker under the framework. Algo-provider arrangements are also governed by exchange requirements.
- API connectivity requires broker-side risk management and auditability. Do not design Hermes to bypass rate limits, tagging, registration or other controls.
- Build a compliance adapter that reads the broker's current API/order rules and exposes them as configuration rather than hard-coded behaviour.

18. Hermes V1 Architecture

- Market Data -> Spot/Futures/Options/L2/Events
- Feature Engine -> Price, volume, OI, IV, Greeks, OFI, microprice, breadth, basis
- Regime Engine -> trend/range/volatility/event states using transparent baselines, HMM/GARCH where useful
- Signal Models -> trend, breakout, pullback, mean reversion, stat-arb, ML
- Options Layer -> surface, IV-RV, skew, gamma estimate, scenario Greeks
- Meta Model -> calibrated P(trade success), expected value after cost
- Risk Engine -> veto authority, sizing, exposure limits, drawdown, kill switch
- Execution Engine -> liquidity-aware order logic, slippage/TCA
- Broker Adapter -> compliant API + audit trail

19. Hermes Training Protocol

- Phase 1: Learn mechanics, statistics and backtesting.
- Phase 2: Build price/volume/basis/OI features and transparent rule-based strategies.
- Phase 3: Add Greeks, IV surface, volatility forecasting, Heston/SABR and scenario analysis.
- Phase 4: Add OFI/MLOFI, microprice, footprint/CVD features and L2 modelling.
- Phase 5: Add XGBoost/LightGBM, calibrated classifiers, then DeepLOB if data support it.
- Phase 6: Add purged/embargoed CV, CPCV, DSR, experiment ledger and model-drift monitoring.
- Phase 7: shadow trading -> paper trading -> tiny live -> controlled scaling.

20. The 'Cheat Code' Stack - Properly Defined

- 1. Information advantage: richer and correctly timestamped data.
- 2. Microstructure advantage: model flow, queues and liquidity—not merely candles.
- 3. Volatility advantage: compare IV with forecast/realized volatility.
- 4. Positioning advantage: estimate OI/GEX/skew while explicitly modelling assumptions.
- 5. Regime advantage: change or disable strategies when market state changes.
- 6. Statistical advantage: calibrated probabilities, expected value and realistic costs.
- 7. Execution advantage: reduce spread, slippage and market impact.
- 8. Risk advantage: survive losing periods and tail events.
- 9. Abstention advantage: trade only when information quality and expected value are sufficient.

21. Final Hermes Rules

- Never assume a research benchmark transfers automatically to NSE.
- Never treat a heuristic threshold as a law of the market.
- Never infer dealer positioning directly from public OI without an explicit assumption model.
- Never evaluate a signal without costs and realistic execution.
- Never tune repeatedly on the final test period.
- Never allow the ML model to override independent risk controls.
- Always preserve a NO-TRADE action.

Source and Verification Register

1. SEBI - Study: Profitability of Individual Traders in the Equity Derivatives Segment (FY25-FY26), Aug 20, 2026: https://www.sebi.gov.in/reports-and-statistics/research/aug-2026/study-profitability-of-individual-traders-in-the-equity-derivatives-segment-fy25-fy26-_103835.html
2. SEBI - Study: Trading Behaviour of Individual Traders in the Equity Derivatives Segment (FY25-FY26), Aug 20, 2026: https://www.sebi.gov.in/reports-and-statistics/research/aug-2026/study-trading-behaviour-of-individual-traders-in-the-equity-derivatives-segment-fy25-fy26-_103836.html
3. NSE - Current Equity Derivatives Contract Specifications, updated Aug 11, 2026: <https://www.nseindia.com/static/products-services/equity-derivatives-contract-specifications>
4. NSE - NIFTY 50 F&O specifications: <https://www.nseindia.com/static/products-services/equity-derivatives-nifty50>
5. NSE Market Pulse - April 2025, including equity-derivatives mode-of-trading statistics: https://nsearchives.nseindia.com/web/sites/default/files/inline-files/Market%20Pulse_April%202025_0.pdf
6. NSE - Retail Algo Trading framework / Threshold OPS: <https://nsearchives.nseindia.com/content/circulars/INVG67858.pdf>
7. NSE Clearing - SPAN and margins: <https://www.nseindia.com/static/products-services/equity-derivatives-nse-clearing-span>
8. Cont, Kukanov & Stoikov - The Price Impact of Order Book Events: <https://arxiv.org/abs/1011.6402>
9. DeepLOB - Deep Convolutional Neural Networks for Limit Order Books: <https://arxiv.org/abs/1808.03668>
10. Moskowitz, Ooi & Pedersen - Time Series Momentum: <https://doi.org/10.1016/j.jfineco.2011.11.003>
11. Heston - A Closed-Form Solution for Options with Stochastic Volatility: <https://doi.org/10.1093/rfs/6.2.327>
12. Almgren & Chriss - Optimal Execution of Portfolio Transactions: <https://doi.org/10.21314/JOR.2001.041>
13. Bailey & López de Prado - The Deflated Sharpe Ratio: <https://doi.org/10.3905/jpm.2014.40.5.094>
14. Soebhag - Option Gamma and Stock Returns, Journal of Empirical Finance (2023): <https://doi.org/10.1016/j.jempfin.2023.101442>
15. Garg et al. - Volatility Risk Premium in Indian Options Prices: <https://doi.org/10.1002/fut.21680>
16. VPIN literature and dispute: Easley/López de Prado/O'Hara, and Andersen/Bondarenko responses. Example: <https://www.sciencedirect.com/science/article/pii/S1386418113000475>

Evidence grading used in this edition

Verified current exchange/regulator: current NSE contract/retail-algo/clearing material and current SEBI study listings. Verified academic phenomenon: peer-reviewed or primary research showing a relationship in a defined dataset. Research candidate: plausible feature or heuristic requiring Hermes-specific validation. Removed/qualified: numerical or causal claims that the source did not support as universal facts.